

Who Knows Better in an Emerging Market? The Performance of Institutions, Foreigners, and Individuals

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Abstract

This paper reconciles previously reported differences in the international finance literature on who invests better in an emerging market. We examine the investment performance of institutions, foreigners, and individuals, using two unique databases of Colombian stocks, as an out-of-sample test of previous findings. We find that local institutions have better outcomes than both foreigners and local individuals on all performance measures. Furthermore, we find that local individuals perform better than foreigners on trading execution, while foreigners generate better long-term returns. Using a bid-ask spread decomposition, we also provide evidence that the better performance of institutions is driven by informational advantages.

JEL: G14, G15, D82

Keywords: bid-ask spread decomposition, foreign investors, individual investors, informed trading, institutional investors.

1. Introduction

Classical asset pricing models assume that agents in financial markets are homogeneous. Market microstructure models, in turn, differentiate between informed and uninformed investors. Such models provide a role for each type of investor in the process of information revealing, and forming prices and liquidity. Accordingly, the literature on investments has devoted a great deal of thought to the differential roles, behavior, and effects of different types of investors. Because of their lower levels of development in terms of liquidity, trading activity, transparency, and efficiency, emerging markets offer particularly fruitful cases for studying the performance of foreigners and locals. It is not clear intuitively whether foreigners or locals are better investors. In one sense, we would expect foreigners to perform better because they have, among other things, deeper pockets, access to more sophisticated research, and greater trading experience. In another sense, a home advantage would give locals access to different and perhaps more valuable sources of information.¹

The empirical literature is ambiguous on the performance and impact of different groups of investors. Several authors find that foreign investors perform better. Grinblatt and Keloharju (2000), for example, find this to be the case in Finland, proposing access to superior information as the probable cause. Richards (2005), studying daily flows to six Asian emerging markets, finds that foreigners enhance their performance basing their net buys on recent local returns. Huang and Shiu (2009) report that Taiwanese stocks with high foreign ownership outperform stocks with lower foreign ownership. The authors argue that this is consistent with foreigners having a greater informational advantage over locals. Ferreira, Matos, and Pereira (2009), in a study that includes twenty-one developed and eight emerging countries, observe that foreign money managers perform

¹ This has been formally proposed in the home-bias literature, e.g., Ahearne, Grier, & Warnock (2004); Nieuwerburgh & Veldkamp (2009).

better than local managers because they have greater access to knowledge, skills, and learning opportunities. Other studies, by contrast, providing evidence that locals have better performance than foreigners. Brennan and Cao (1997), for example, propose a model in which locals have an informational advantage and find support for it in portfolio investment flows between the United States and four other developed countries. Hau (2001) compares the performance of German stocks by professional traders located in and outside Germany, again showing that locals produce better investment returns. In emerging markets, a number of recent papers show that locals perform better (e.g., on Indonesia, see Agarwal, Faircloth, Liu, & Rhee, 2009; Dvořák, 2005; on China, see Chan, Menkveld, & Yang, 2008; and on Korea, see Choe, Kho, & Stulz, 2005). Moreover, in the context of financial crises, several studies report that locals anticipated crashes better than foreigners (see, e.g., Kaufman, Mehrez, & Schmuckler, 2005; Kim & Wei, 2002).

This literature review indicates that evidence on performance is mixed. However, our view is that, instead, the debate on locals versus foreigners and institutional versus individual investors has been misrepresented. We believe that the differences in these findings stem from imprecise definitions of investor groups and insufficient acknowledgment of the differences among performance measures. If the performance results and the investor categories are described more precisely, many of what at first appear to be contradictions in the literature disappear. Posing the research question as “foreigners versus locals” is not fair or appropriate because most foreigners reported in previous studies are actually institutions, so comparing their performance with that of the entire local group, which comprises both households and institutions, is not a comparison of equals. Moreover, even “locals” means different things in a developed stock market than in an emerging one. In developed markets, domestic investors normally used to proxy locals heavily comprises institutions, whereas in emerging markets it has a greater share of individuals than

institutions. For example, local individuals account for 60-75% of the value traded in Korea (Choe et al., 2005) and 90% in Taiwan (Barber, Lee, Liu, & Odean, 2009). In the database used in our study, the size of local individual trading is not quite as large as in Asia, but it is still significantly larger than in developed markets. Colombian individual investors account for 36% of the value traded, with institutions comprising 52% and foreigners the remaining 11%. In a typical developed market, the size of the local investor group is usually much smaller. In a sample of stocks traded on the New York Stock Exchange (NYSE), Kaniel, Saar, and Titman (2008) report that individuals account for about 4% of the weekly value traded. For these reasons, the traditional debate pitting foreigners against locals is misleading. More meaningful comparisons are, on the one hand, local institutions versus local individuals and, on the other hand, local institutions versus foreign institutions. This is the approach adopted here, and the performance literature would benefit from doing the same.

The following studies compare performance among the three groups of investors, although using different approaches and consequently finding contradictory results. Choe et al. (2005) find in Korea that domestic money managers obtain better execution prices than foreign institutions and that domestic individuals trade at better prices than foreign institutions. They discard explanations of price impact and information, instead offering return chasing by foreigners as a possible reason. Using transaction data from Taiwan, Barber et al. (2009) find, on the contrary, that individual investors systematically underperform both local and foreign institutions. Local institutions have the best performance of the three groups in both aggressive and passive trades. Finally, Lee, Liu, Roll, & Subrahmanyam (2004), using intraday data from Taiwan, study the order imbalance and net order flow of the three types of investors. They find that large domestic institutions are

responsible for most of the informed trading while large individuals tend to be noise traders or liquidity providers.

We make two contributions to the literature. First, we replicate and verify several of the results of the existing literature in another country, Colombia. By estimating different performance measures according to the type of investor, we conduct an out-of-sample test of the most important results identified in the existing literature, and reconcile the mixed results mentioned earlier. Our study makes use of a unique and detailed database from the Colombian Stock Exchange (BVC) that compiles buys and sells by different types of investors for every Colombian stock on a daily basis, spanning more than seven years. Investors are categorized as foreign, individual, or institutional, with institutional investors broken down into various subcategories. To our knowledge, this is the first study of its type conducted in a Latin American market.

Second, we go beyond the previous literature by providing evidence of a link between short-term performance and having an informational advantage. Market microstructure models, such as Hasbrouck (1991) and Kyle (1985), formalize a relationship between informed trading and illiquidity, usually measured as increasing bid-ask spreads. Other studies provide supporting evidence of this relationship (see, e.g., Easley, Engle, O'Hara, & Wu, 2008; Easley, Hvidkjaer, & O'Hara, 2010; Lei & Wu, 2005), including one Latin American study that incorporates Colombian stock data (AUTHOR 2015). Some of those models imply that superior information should move prices in the direction of the information (Easley & O'Hara, 1992; Glosten & Milgrom, 1985; Kyle, 1985). Furthermore, several studies propose methodologies that extract the bid-ask spread component related to adverse selection (Huang & Stoll, 1997; de Jong, Nijman, & Röell, 1996; Madhavan, Richardson, & Roomans, 1997). This paper, starting with trade and quote data, uses the spread decomposition model of Madhavan, Richardson, and Roomans (1997) (henceforth,

MRR) to estimate the adverse selection component of the bid-ask spread for Colombian stocks during a period of two years. This serves as a proxy for informed trading and enables us to test for a direct relationship of the adverse selection component with the trading intensity of each group of investors. If these results align with those of performance at daily frequency, they will provide evidence that superior short-term performance can be explained at least partially by an informational advantage.

Our main results can be summarized as follows. First, we provide evidence that both local individuals and institutions have an execution edge over foreigners, by trading at more favorable prices. This is consistent with the findings of Choe et al. (2005) in Korea. Moreover, different performance measures reveal that, over the long term, local institutions are more effective traders than foreigners, who in turn are more effective traders than local individuals. In the short term, the results show that institutions have the upper hand. In addition, we provide evidence to explain the short-term edge in terms of informational advantage. Specifically, after applying the MRR spread decomposition model, we find a positive relationship between institutional trading and the adverse selection component of the bid-ask spread. The corresponding effect is mixed for individuals and negative for foreigners. This evidence appears to suggest that institutions, and to some extent individuals, are exploiting a short-term informational advantage.

Additionally, we investigate how this institutional edge differs between long-term local institutions (e.g., pension funds) and short-term ones (e.g., brokerage firms), by comparing their short- and long-term performance. In principle, we assume that each group has a comparative advantage in its corresponding investment span. We find that long-term local institutions have better one-year performance than short-term local institutions, as expected, due more to the stocks they sell or avoid buying than to the stocks they buy most. In contrast, we find that short-term

institutions have only a weak advantage on a daily basis, explained by their better relative trade execution but weakened by poorer daily timing. We also find no significant difference in their one-month performance. These results differ from Yan and Zhang (2009), who reports that in the United States short-term institutions are better predictors of prices and that this advantage does not disappear in the long run.

Finally, our results reconcile the apparently contradictory findings between previous studies focused on emerging markets, which indicate a local advantage (Agarwal et al., 2009; Brennan & Cao, 1997; Dvořák, 2005), and those that conclude the opposite (Huang & Shiu, 2009; Richards, 2005). The contradictions have two origins. First, within the local grouping, previous authors do not separate institutions from retail traders. Specifically, studies such as Agarwal et al. (2009) and Dvořák (2005) do not differentiate local trades between these groups. Brennan and Cao (1997) and Richards (2005) only observe foreign flows. Huang and Shiu (2009) do not differentiate ownership by the two local groups. Second, previous studies use different performance methodologies. For example, we reconcile the apparently contradictory results of Choe et al. (2005), who report local individuals have an advantage over foreigners, with those of Barber et al. (2009), who suggest the opposite. The difference occurs because Choe et al. (2005) measure trade execution, whereas Barber et al. (2009) compare long-term performance. We are able to replicate both results and find that both conclusions are valid and entirely consistent with each other.

The two studies most closely related to our paper are Barber et al. (2009) in Taiwan and Choe et al. (2005) in Korea. Both compare the performance of the three types of investors. Our study varies from those papers in at least two respects. First, by applying a spread decomposition model, we provide evidence of a relationship between informed trading and superior short-term performance among local institutions. This relationship is consistent with the predictions of the

different market microstructure models (Glosten and Milgrom 1985; Kyle 1985) and with previous empirical evidence on the effect of informed trading on prices (see, e.g., Easley, Engle, O'Hara, & Wu, 2008; Easley, Hvidkjaer, & O'Hara, 2010; Lei & Wu, 2005). Second, our study includes both trade execution and performance by the different types of traders and measures them both separately and jointly. The previous two studies focus only on one of them. This is not a minor detail, since the sizable average execution advantage that locals have over foreigners, as reported by Choe et al. (2005), could explain the difference in their performance.

Our paper is organized as follows: Section 2 describes the two unique databases used in this study. Section 3 describes the methodology, with details on the different measures of performance employed. Section 4 presents and discusses the results of differential performance in the short and long term and the differential effects on informed trading. Finally, section 5 concludes.

2. Data

We take advantage of a unique database from the Colombian Stock Exchange (BVC), as described in Table 1. This database summarizes information on daily buys and sells by different types of investors for each stock traded on the BVC from January 1, 2007, to May 15, 2014. Buys and sells are measured in number of operations, traded value, and traded volume. Type of investors include local institutions, foreigners and local individuals. Local institutions include brokerage firms, pension funds, mutual funds, trusts, family offices, banks, and other financial institutions. For most of our purposes, we group together all types of local institutional investors. However, we performed some additional analysis differentiating between short-term institutions, that is, brokerage firms and family offices, and long-term institutions, namely, pension funds, mutual

funds and trusts. From the stock exchange, and confirmed by two major Colombian brokerage houses, we know that only institutions are included in the foreign group, mostly international asset managers and mutual funds, aside from some hedge funds and a few foreign direct investments.² For each stock day and each type of investor, the database reports value weighted average price, traded volume (number of stocks), traded value in US dollars and number of operations, separately for buys and sells. This information is summarized in Table 1 by quartiles of market capitalization. The BVC generates this database from the stock exchange transaction, that include identification for both parties in each trade and allows then to classify both in the various groupings based on the unique tax number (NIT).

Table 1 shows some interesting differences. In the full sample, institutional investors do most of the trading in terms of money, 53% in the full sample, compared with 11% for foreigners and 36% for individuals. Institutions are responsible for more than half the traded value for the three top size quartiles, surpassed only by individuals in the bottom quartile. This higher role of individuals in trading smaller stocks is consistent with the US market (Barber, Odean, & Zhu, 2008). Foreign investors, in turn, have the smallest share of the traded value in all quartiles, which coincides with those reported for Indonesia (Dvořák, 2005), Korea (Choe et al., 2005), and Taiwan (Barber, et al., 2006). When trading activity is measured by number of trades, institutions and individuals change places. In this case, individuals are responsible for more than half the trades in every quartile, as much as 68% in the bottom quartiles. This pattern is similar to the one reported in the studies mentioned earlier about Indonesia, Korea, and Taiwan. A familiar pattern also emerges in estimates of the daily average trading size. Institutional investors have the largest trades on average, closely followed by foreigners, for the full sample and in each size quartile except the

² We thank Jairo Julian Agudelo from Valores Bancolombia, a leading Colombian brokerage house, for this insight.

lowest one. Individual investors in contrast present average trade sizes two to three times smaller than the other two groups.

We perform the following adjustments to the data. First, in the original BVC database we verify the balance between stocks bought and sold on each stock day. We do this for traded value, the number of shares, and the number of operations. We find only very minimum discrepancies between the total sell and total buys balances per stock day. Second, because the BVC database reports historical prices, we calculate the adjustment factors required to properly account for splits and dividends paid in stocks. Third, we decide not to take into account the trading activity of the American Depository Receipts (ADR) group. Colombia has four stocks listed as ADRs type II or III, all of them on the NYSE. Most trades classified as such are either part of arbitrage operations between the underlying local stock and the ADR itself or intended to move capital between the two exchanges. Since we are unable to classify those trades as belonging to institutions, foreigners, or locals, we omit them from our analysis. Those trades are about 4.75% of the overall trading value in the sample.

In order to test for informed trading in a market microstructure setting, we take a trade and quote database for Colombian stocks from Bloomberg, for the period August 23, 2010, to August 31, 2012.³ This database is made up of the time-stamped trades and quote revisions for the 42 most traded stocks in Colombia. Stocks on the BVC trade in a pure limit order book market, without designated market makers or dealers. Furthermore, the market operates most of the time as a continuous market but closes in a call auction that lasts approximately five minutes. An additional feature of the Colombian stock market are circuit breakers that from time to time interrupt the

³ Bloomberg stores only six months of trade and quote data, so this database had to be hand collected over a period of two years.

continuous market (“volatility call auctions”). Circuit breakers are triggered by sharp price movements and last a few minutes. Because spread decomposition models are based on a continuous market, we remove the trades and quote revisions that correspond to both closing and volatility call auctions.⁴ After cleaning up the data, we end up with a total of approximately 740,000 trades and around 3 million quote revisions. To implement the MRR decomposition model as applied in Ahn, Cai, Hamao, and Ho (2002), we classify each trade as a buy or sell initiated using the algorithm of Lee and Ready (1991).

3. Methodology

3.1. Performance measures by investor group

We measure the performance of the different types of investors in three dimensions: trade execution, short-term performance, and long-term performance.

To measure trade execution, we estimate the average execution cost separately for buys and sells. Specifically, we calculate the proportional difference for each type of investor j of the value-weighted average trading price for buys (sells) with the value weighted average daily price (VWAP) of the stock i on day t as follows:

$$Average_buy_cost_{ijt} = (VWA_buy_price_{ijt} - VWA_price_{it}) / VWA_price_{it} \quad (1)$$

$$Average_sell_cost_{ijt} = (VWA_price_{it} - VWA_sell_price_{ijt}) / VWA_price_{it} \quad (2)$$

⁴ These are identified from a database provided by the BVC.

Because we account for both sides of each trade on any given stock day (with the noted exception of ADRs), average costs should add up to close to zero among the type of investors. A positive average buy or sell cost can be interpreted as aggressive trading—for example, posting mostly market orders, creating price impact by demanding liquidity beyond the bid/ask depth, or acting as intraday momentum traders. To some extent, this could also be due to prices' moving against the trade, as in Choe, Kho, and Stulz (2005). Conversely, a negative average buy or sell cost means less aggressive trading, which can include supplying liquidity with limit orders, order execution strategies that minimize price impact, or acting as intraday reversal traders.

Short-term (daily) performance is measured in three ways. First, a very simple measure takes the next-day return of stock i (the negative of it) if the type of investor j had a positive (negative) net total buy on the current day t , as follows:

$$Next_day_return_{ijt} = sign(Net_buy_{ijt}) \times \left(\frac{VWA_price_{it+1}}{VWA_price_{it}} - 1 \right) \quad (3)$$

Where: $Net_buy_{ijt} = Q_buy_{ijt} - Q_sell_{ijt}$, this is the net buy is defined as the difference in quantities bought and sold by the type of investor on the current day.

Whereas equation (3) is a straightforward measure of short-term performance, it does not take into account the magnitude of the net buy. We propose a net profit measured, roughly following Dvořák (2005), based on the next-day increase in value of the position of a type of investor j due to the net buy on day t and the price increase in $t + 1$. We normalize it with the average traded value per type of investor, as follows:

$$Net_profit1_{ijt} = \frac{(Q_buy_{ijt} - Q_sell_{ijt}) \times (VWA_price_{it+1} - VWA_price_{it})}{Average_trading_value_{ij}} \quad (4)$$

Where: $Average_trading_value_{ij} = \frac{\sum_{t=1}^T (Q_buy_{ijt} \times VWA_buy_price_{ijt} + Q_sell_{ijt} \times VWA_sell_price_{ijt})}{T}$

Equation (4) captures the profit from both buying a stock that increases in price at the next day and selling a stock that decreases. However, it does not incorporate the effect of differential trade execution among the three types, because it assumes that buys and sells are made at the same daily price, VWA_price_{it} . In response, we propose the following alternative measure that takes into account the specific execution prices by type of investor:

$$Net_profit2_{ijt} = \frac{Q_buy_{ijt} \times (VWA_price_{it+1} - VWA_buy_price_{ijt}) - Q_sell_{ijt} \times (VWA_price_{it+1} - VWA_sell_price_{ijt})}{Average_trading_value_{ij}} \quad (5)$$

As a result, equation (5) incorporates both the short-term performance of equation (4) and the execution cost effect of equation (1).

The measures presented above focus on one-day performance, which might be related only to short-lived information. In order to measure monthly performance, we form portfolios of stocks that reflect the preferences of institutions, foreigners, and individuals, loosely following Barber, Odean, and Zhu (2008). Specifically, at the end of each month we rank stocks depending on the monthly net buy for each type of investor and form equally weighted portfolios accordingly. For instance, portfolio $Q4_netbuy_{jt}$ ($Q1_netbuy_{jt}$) is an equally weighted portfolio of the stocks most bought (sold) by type of investor j in month t . We estimate the return of this portfolio in month $t + 1$ and rebalance it at the end of month $t + 1$. Thus, these portfolios represent a way to implement a strategy that uses the information content of net buys by type of investors and to measure performance. At the same time, it is expected that $Q4_netbuy_{jt}$ outperforms the corresponding

$Q1_netbuy_{jt}$ if the type of trader j has any stock-picking abilities. Their returns provide a measure and comparison of the overall stock-picking abilities of the different type of investors.

In a similar fashion, to estimate the one-year performance, we take the same portfolios based on monthly net buy by investor, described above, and estimate their return in the twelve-month period following the month, with no rebalancing during that twelve-month period. Because of overlapping between portfolios within a twelve-month span, they cannot be considered returns from the same strategy, unlike the one-month returns. Anyway, the raw returns of those portfolios provide a way to measure the long-term performance of each type of investor.

3.2. Informed trading effects by investor group

As mentioned above, we search for evidence of the link between information and performance. In market microstructure models, informed trading can be detected by its effects on the bid-ask spread (Easley & Hara, 1992; Easley et al., 2008; Lei & Wu, 2005; AUTHOR 2015). We use the MRR model to conduct a decomposition of the spread in the trade and quote database of Colombian stocks mentioned above. This model has previously been used in electronic order-book markets, as in Japan (Ahn et al., 2002) and the Korean Stock Exchange (Eom, Ok, & Park, 2007). Specifically the MRR decomposition model, as implemented in Ahn et al. (2002), estimates the implicit bid-ask spread components based on the change on transaction prices, in the following model:

$$\Delta P_{\tau} = \alpha(Q_{\tau} - \rho Q_{\tau-1}) + \beta(Q_{\tau} - Q_{\tau-1}) + u_0 + u_{\tau} \quad (6)$$

where ΔP_τ is the change in consecutive transaction prices, Q_τ is a buy-sell trade identifier, equal to 1 (-1) if the trade was buy initiated (sell initiated). In turn, u_0 is a constant drift, and u_τ is the disturbance term. Parameter ρ is the autocorrelation between identifiers of successive trades, rendering $\rho Q_{\tau-1}$ as the expected sign of transaction τ . In turn, α measures the part of the bid-ask spread associated to the revision in expectations, this is, the adverse selection component, and β , is the change in price due to bid-ask bounce, revealing the component of the spread associated to liquidity supply: the inventory and order processing costs. Thus $\alpha + \beta$ is the estimation of the implied spread. Equation (6) is estimated using a GMM procedure.⁵

Because we are interested in the time variation of the spread components to relate it to the trading activity of the different investor groups, we estimate equation (6) on a stock-day basis. Because of insufficient trading activity, the model cannot be estimated in all cases. In the end, the model was estimated using a sample of the six most actively traded stocks, totaling 2,121 stock days.

To test whether particular types of investors are associated with increased informed trading, we regress the estimated implied spread ($\alpha + \beta$) and the adverse selection component (α), both scaled by the average trading price on the stock day, against measures of trading intensity of the two groups, controlling for known determinants of the spread. The base model is the following:

$$\begin{aligned} \text{Implied_spread}_{it} = & \text{Volat}_{it} + \text{Log_num_op}_{it} + \text{return}_{it} + \text{Log_Price}_{it} + \\ & \text{Log_Avg_trade_Size}_{it} + \text{Log_|OrderImbal|}_{it} \end{aligned} \quad (7)$$

⁵ The GMM procedure is estimated with the following four moment restrictions: orthogonality between the revision in the identifier and its previous value: $Q_{\tau-1}(Q_\tau - \rho Q_{\tau-1}) = 0$; the zero mean of the disturbance: $u_\tau - u_0 = 0$; and the orthogonality conditions between the disturbance and the identifiers on τ and on $\tau - 1$: $Q_\tau(u_\tau - u_0) = 0$, $Q_{\tau-1}(u_\tau - u_0) = 0$.

$$Adv_selec_comp_{it} = Volat_{it} + Log_num_op_{it} + return_{it} + Log_Price_{it} + Log_Avg_trade_Size_{it} + Log_|OrderImbal|_{it} \quad (8)$$

Following similar models in the literature (e.g., De Cesari, Espenlaub, & Khurshed, 2011; Grullon, Kanatas, & Weston, 2004; Hendershott & Moulton, 2011), we include as control variables the volatility of the stock as log of the ratio between the maximum and minimum prices during the day ($Volat_{it}$), the log of the number of operations as a measure of trading activity ($Log_num_op_{it}$), the return on the stock ($return_{it}$), the log of the price (Log_Price_{it}), and the log of the average trade size $Log_Avg_trade_Size_{it}$. We include the log of the absolute value of the order imbalance, $Log_|OrderImbal|_{it}$. The order imbalance is the difference between buy initiated minus sell initiated trades and is a known determinant of the bid-ask spread. The classical market microstructure models posit that the order imbalance is in part a result of the activity of informed traders (Easley & O'Hara, 1992; Glosten & Milgrom, 1985; Kyle, 1985). Empirically, order imbalance has been shown to predict very short-term returns (Chordia & Subrahmanyam, 2004), and its magnitude has a positive relation with measures of illiquidity, such as the proportional bid-ask spread (Chordia, Roll, & Subrahmanyam, 2002) and the price impact (Lee et al., 2004).⁶

We adapt this base model to test for the differential role of the three types of investors in the implied and asymmetric spreads. Because the trade and quote data from Bloomberg lack any identification of the type of investor in each trade, we cannot precisely measure the order imbalance from each type, and we have to use proxies. Specifically, we run alternative versions of equations

⁶ Easley et al. (2008) propose the Dynamic PIN model as a measure of the probability of informed trading, which can be seen as a transformation of the absolute value of the order imbalance. They provide evidence of the positive relation of the Dynamic PIN and the bid-ask spread. In turn, AUTHOR (2015) provide similar evidence for six Latin American stock markets, including that of Colombia.

(7) and (8), replacing $\text{Log_}|\text{OrderImbal}|_{it}$ with two proxies of the order imbalance brought by each type of investor. First we use the interactive variable $\text{Log_}|\text{OrderImbal}|_{it} \times \text{Share_trading}_{ijt}$, the product between the order imbalance variable and the share of trading by type of investor (measured by number of trades). Second, we use $\text{Log_}|\text{Net_buy}_{ijt}|$, a transformation of the net buys by type of investor, estimated in turn as the difference between the numbers of buys and the number of sells.⁷

4. Results

4.1. Differential short-term performance

We start by investigating the differential short-term performance of the three types of investors: local institutions, foreigners, and individuals. Table 2, Panel A, presents the results of the average trading cost in both buys and sells by the three types of investors, for the full sample and four quartiles of size. Interestingly, individuals achieve the best execution of all, with statistically significant negative average cost all across the quartiles and particularly large savings in purchases, with an average cost of -0.1%. Institutions are second, also with significantly negative average costs in buys and sells, the largest average savings in sells, and significantly negative selling cost for each size quartile. Local institutions' cost savings disappear only for the buys in the two bottom size quartiles. In stark contrast, buys and sell by foreigners have the largest average cost for every size quartile. Since by definition measures (1) and (2) are zero-sum on each stock day, we can say that average savings by institutions and individuals occur at the expense of foreigners. Panel B

⁷ To be clear, the buys (sells) in the BVC database lump together both buy-initiated and sell-initiated buys (sells), whereas the order imbalance is the difference between buy-initiated and sell-initiated trades.

presents the results of Wilcoxon tests comparing distributions of trading cost across types of investors. Institutions appear to be more efficient sellers than both foreigners and individuals in the full sample and every quartile. Individuals, in turn, consistently show the lowest trading cost for buying of the three types. Finally, foreign investors are the least-efficient traders at both buying and selling in all five groupings of the data.

Taken together, these findings, along with those of average trading size in Table 1, in the absence of more detailed data, are consistent with foreigners posting large trades at a disadvantage with locals. In turn, individuals appear to have a smaller price impact (and perhaps more of the liquidity providing), and institutions, a more efficient trading execution, especially of sales, in spite of their large average trades.⁸ These results can be interpreted as individuals and institutions being more cautious traders than foreigners and coincide with the findings of Agarwal et al. (2009) for Indonesia and Choe et al. (2005) for Korea.

Table 3 presents the results of the simple daily performance measure (3) for the three types of investors, for the full sample and the four quartiles of size. None of the three types have a daily performance significantly different from zero for the full sample. However, both institutions and foreigners have a positive and statistically significant performance in the top size quartile, and foreigners also have it in the third quartile. In contrast, individuals have a negative performance in the two top quartiles of size. Because daily average returns for stocks are not significantly different from zero, average performance (equation 3) should be around zero each stock day, and positive performance by institutions and foreigners is linked to a negative one by individuals. Thus, these findings suggest that, for the larger stocks, institutions and foreigners tend to buy (sell) before next-

⁸ On the other hand, we know anecdotally that foreigners tend to trade at the closing call auction, to make sure they match the closing price, which in turn seems to make them trade at disadvantage. However, with the available data we are unable to provide evidence for either claim.

day stock price rises (falls), whereas individuals do the opposite. Table 3 also presents the results of Wilcoxon tests that compare the distributions of performance across types. Overall, institutional investors demonstrate significantly better timing than both foreigners and individuals, in almost every size quartile.⁹ Foreigners, in turn, beat individuals in all five cases. In principle, this is consistent with an informational advantage for institutions relative to the other two types. Yet this is only an indicative result because measure (3) is too simple to capture the diverse aspects of performance.

Net profit measures (4 and 5), unlike (3), takes into account the net buy of each of the three investors on day t and the return on day $t + 1$, with equation (5) also incorporating the price execution on day t . The results of net profit 1 (equation 4) are presented in Table 4, portraying superior performance by institutional investors in the full sample, with positive and statistically significant net profit as well as in quartiles 4 and 2, and positive performance, which is marginally significant, in the lowest quartile. Foreigners, in turn, have overall performance that is not statistically different from zero, with the only exception of a positive net profit measure in the third quartile. Individuals, in turn, have significant negative performance overall and in every quartile. The results of Wilcoxon tests comparing performance across types show the same ranking as in Table 3: Institutions significantly beat both types in each size decile (the only exception is foreigners in the third quartile), and individuals have the worst performance. Overall, the outcome of Table 4 is similar to that of the simple daily measure (equation 3) in Table 3, with individuals trading at a disadvantage with local and foreign institutions, as reported in Taiwan (Barber et al., 2009).

⁹ The only exception is the performance of foreigners in the third quartile, which we verified, is not due to outliers or any single specific stock.

The results of net profit 2 (equation 5) are presented in Table 5. Net profit 2 can be seen as the most complete performance measure. It is a refinement of the daily performance measure net profit 1, which incorporates the trade execution cost (equations 1 and 2) presented in Table 2. As expected, the average performance of institutions in the full sample is superior to the one in Table 4 (0.04% vs. 0.03%), associated with their negative execution costs both at buys and sells in Table 2 (-0.01% and -0.03%). Moreover, this positive performance by institutions is statistically significant for the full sample and the top three size quartiles. Foreigners in turn have significantly negative performance in the full sample (-0.27%), in contrast to that of net profit 1 (0.03%) but clearly explained by the execution cost in both buys and sales reported in Table 2 (0.09% and 0.07%). Finally, individuals have a negative overall performance (-0.02%), slightly inferior to the one in Table 4 (-0.03%), due to their negative execution cost in Table 2 (-0.01% for buys and -0.03% for sells). Still, individuals have negative and statistically significant performance overall and in the quartiles 4 and 2. The Wilcoxon tests imply that institutions have overall daily performance superior to that of the other two types, in all cases, with the only exception being the smallest stocks, in which there is no statistical difference with individuals. Individuals, in turn, do perform better than foreigners in the full sample and in three size quartiles.

Taken together, the results in Tables 3 to 5 show that local institutions have better daily performance than the other two types, tending to execute larger net buys (sells) in stocks with positive (return) at the next day and benefit from better trade execution. In turn, foreigners and individuals have different drawbacks. Foreigners have performance before costs no different from zero, along with market timing better than that of individuals. However, their overall performance turns negative, and lower than that of individuals, because of their larger trading costs. Finally individuals, have the worst timing of the three types, tending to have larger net buys (sells) in stocks

with negative (positive) return in the following day, but their performance is helped by their favorable execution cost, ending up better than that of foreigners. This differential short-term performance of the three types might be related to an informational advantage, which will be tested in a market microstructure model.

4.2 Differential long-term performance

Table 6 summarizes the performance of the portfolios of the stocks most bought ($Q4_netbuy_{jt}$) and stocks most sold ($Q1_netbuy_{jt}$) by each of the three type of investors, estimated from the monthly total net buy and recomposed monthly, as described in the Methodology section. We also include a $Q4 - Q1_netbuy_{jt}$ portfolio, long in the most bought stocks and short in the most sold ones. Panel A presents the summary statistics of the monthly returns and compares them with returns of the corresponding portfolios for the other types. Panel B shows the alpha and beta from three-equation SUR models for a market index model, one for each type of portfolio. These three-equation SUR models regress the portfolio excess return, for each of the three types of investors, against the excess return of the COLCAP. Finally, Panel C shows the summary statistics of the twelve-month returns of the portfolios and the statistical comparison the corresponding portfolios of the other types.

Panel A of Table 6 indicates that the portfolios $Q4_netbuy_{jt}$ and $Q4 - Q1_netbuy_{jt}$ of institutions have returns that are positive and statistically significant at the 10% level, pointing to better performance by those investors. For foreigners and individuals, only the $Q1_netbuy_{jt}$ portfolio delivers returns that are statistically significant but positive, suggesting deficient stock picking by those investors, since the stocks they sell the most have positive returns on average,

even above those they buy most, as implied by the negative average returns of the $Q4 - Q1_netbuy_{jt}$ portfolios. Moreover, the results of the t test in Panel A show that the institutional $Q1_netbuy_{jt}$ portfolio renders a statistically lower return than those for foreigners and individuals. Additionally, the institutional $Q4 - Q1_netbuy_{jt}$ portfolio has a higher average return than those for the other two types. Consistently, the results in Panel A indicate better one-month performance for institutions and worse performance for foreigners and individuals.

Of course, the better stock picking of institutions over the other two types might well be due to differences in systemic risk. However, the estimations of Jensen's alpha presented in Panel B reveals that this is not the case. Institutions hold the only significantly positive alpha for the $Q4 - Q1_netbuy_{jt}$ portfolio, the only positive alpha for the $Q4_netbuy_{jt}$ portfolio, and lower for the $Q1_netbuy_{jt}$ portfolio. Moreover, the results of a Wald test show that the alphas for both institutions and foreigners in the $Q1_netbuy_{jt}$ portfolio are statistically significantly lower than for individuals.

So far we have found that institutions consistently beat both foreigners and individuals at daily and monthly frequency. But is this advantage short lived? To answer that question, we estimate and compare the twelve-month average return of the portfolios, as presented in Panel C of Table 6. Interestingly, $Q4_netbuy_{jt}$ portfolios for institutional and foreign investors deliver positive and highly significant twelve-month returns, both of them statistically higher than that of individuals. In turn, the $Q1_netbuy_{jt}$ portfolios for institutions and foreigners deliver returns that are significantly below those for individuals. Moreover, the $Q4 - Q1_netbuy_{jt}$ portfolio for institutional investors deliver statistically higher returns than those for foreigners and that in turn are higher than those for individuals. Overall, those findings reflect better one-year performance

for the stocks preferred by institutions, intermediate performance for foreigners, and worse performance for individuals.

To summarize the results of Table 6, institutional investors as a group show the best ability at choosing stocks compared to foreigners and individuals, measuring performance in periods of one and twelve months. This result is driven more by the stocks that institutions avoid or sell the most than by those they tend to prefer. In turn, foreigners, while showing comparable ability to individuals, when performance is measured in monthly frequency, deliver better performance when measured over one year periods. These results are similar to those obtained in previous studies comparing the performance of the same three groups of investors in Taiwan by Barber et al. (2009). The authors argue that the lower performance of individuals can be explained by aggressive orders. In contrast, our results point to inferior stock-picking ability, especially in the stocks they sell the most.¹⁰

4.3. Differential informed trading effects

Here, we explore whether the variation in short-term performance across investor groups can be related to differentials in informed trading effects. As described above, the MRR model offers a way to extract the adverse selection cost component of the bid-ask spread from order processing costs. If a particular group of investors had an informational advantage, that should be reflected in a higher adverse selection cost component in days when they trade the most. We are able to test

¹⁰ As a robustness test, we also run the analysis of Table 6 with value-weighted portfolios, instead of equally weighted, using the market capitalization of each firm. The unreported results, available upon request from the authors, are qualitatively similar, but show foreigners with comparable performance to local institutions for both the one-month and the twelve-month periods. This is explained by the huge weight of the company Ecopetrol in value-weighted portfolios for the BVC. In the sample period, Ecopetrol comprised between 40% and 60% of total market value.

this hypothesis by estimating equations (7) and (8) on the implied spread and the adverse selection component of the spread, calculated with the MRR model and scaled with the average trading price, controlling for other known determinants of the bid-ask spread.

Table 7 presents the results of fixed-effects panel data models that regress the implied spread and its adverse selection component, estimated on a stock-day basis, against order imbalance measures and determinants of the bid-ask spread. In columns 1 and 2, we present the results of the base models (equations 7 and 8).¹¹ As expected, we obtain a highly significant relation between the implied spread and trading activity ($Log_num_op_{it}$), volatility ($Volat_{it}$), and the log of price (log_P_{it}), with the expected sign, as previously reported in similar models (AUTHOR 2015; De Cesari et al., 2011; Grullon et al., 2004). Most important, the absolute value of the order imbalance, a proxy of informed trading activity, has a positive coefficient in the regression of the adverse selection component of the bid-ask spread (column 2), as expected.

In the columns 4 to 6, we replace the order imbalance variable with proxies for the order imbalance by investor, as described in the Methodology section. In column 4, using $Log_|OrderImbal|_{it} \times Share_trading_{ijt}$ we obtain a negative and highly significant effect from foreigners on the adverse selection component of the spread, a positive and significant effect from institutions, and a positive for individuals, but only significant at the 10% level. A Wald test validates that the effect from foreigners is significantly lower than for the other two types, but they are not significant between them. In column 6, using $Log_|Net_buy_{ijt}|$, we again find a negative and highly significant coefficient for the order imbalance proxy for foreigners and positive

¹¹ We run the panel data models using Driscoll and Kraay's covariance matrix estimator (Driscoll & Kraay, 1998), which provides a heteroskedasticity and autocorrelation consistent covariance matrix (Hoechle, 2007). In addition, this approach is consistent independently with the relative panel cross-sectional and temporal dimensions.

coefficient for institutions, significant at the 5% level. Individuals have a negative effect, which is marginally significant. When the coefficients are compared using a Wald test, the effect for institutions is significantly higher and the one for foreigners significantly lower, than the other two.

All in all, the results in Table 7 support the hypothesis that the superior short-term performance for institutions, demonstrated in the short-term performance measures reported in Tables 3 to 5, can be related to an informational advantage. The nature of this short-term information is not clear, however, Leaño and Pedraza (2016) have provided evidence of an informational advantage for a particular class of institutions in Colombia: pension funds, when trading the stocks of companies from the same group as they belong to. In turn, there is some evidence of foreigners trading with an informational disadvantage in the short term, even when compared to individuals, which goes along with the result of the net buy 2 measure in Table 5, which measures short-term performance for the full sample.

4.4. Short-term vs. long-term local institutions

We take advantage of the detail in the classification of local institutional investors in the BVC database to compare the performance of short-term institutions vs. long-term ones. Specifically, we classify brokerage firms (executing their own trades) and family offices as short-term institutions. Anecdotally, we know that these two are very active traders on the Colombian stock market, and the summary statistics confirm this observation. As presented in Table 8, these two types of institutions account for 61% of the total value traded and 69% of the operations by institutions. In addition, consistent with more speculative behavior, their share of trading activity is larger among smaller firms. They have 69% of the total value traded and 79% of the operations

in the smallest size quartile. Finally, some long-term institutions, such as pension funds, insurance companies, and investment companies, have an average traded value per operation well above that for short-term institutions.

We group pension funds, mutual funds, insurance companies, investment companies, and investment trusts together as long-term institutions. As expected from their investment style, they conduct a minor but still sizable share of the trading activity: 39% of the traded value and 27% of the operations. We exclude banks and other financial intermediaries in both groups, because their investment style and span are not clear. By doing so, we exclude only 0.4% (0.3%) of the trading activity by local institutions measured as the value traded (number of operations).

Contrasting the daily performance of the two groups of local institutions, we find an interesting story. The results of the simple measure in equation (3), in Table 9, Panel A, show negative performance for short-term institutions in the full sample and the top two size quartiles, with a statistically significant difference in favor of long-term institutions. Panel B, reporting net profit (equation 2), shows a more equilibrated picture, with short-term institutions having positive and statistically significant performance in the full sample and the three top quartiles. However, the average performance of short-term institutions is statistically better than that of long-term institutions only in quartile 2. The difference in results between Panels A and B of Table 9 can be explained by the average trading cost. In unreported results, we find that both groups have negative average trading costs, consistent with those in Table 2, but those of the short-term group are statistically smaller than in the long term for both buys and sells in the full sample. Thus, short-term institutions have an edge in terms of execution over long-term ones, which appears consistent

with their larger average trading size in Table 8.¹² However, the timing of trading of short-term institutions is worse (Table 1) or at least not better than that of their long-term peers, and this is confirmed by omitted results of the net profit 1 measure (omitted). Overall, the superiority of short-term institutions in short-term performance over long-term ones is tenuous, at best.¹³

In contrast, long-term institutions do have an edge in long-term performance, as expected. Table 10 presents the performance of portfolios of the most-bought stocks ($Q4_netbuy_{jt}$) and the most-sold stocks ($Q1_netbuy_{jt}$) for each of the two groups of institutions, as in section 4.2 and Table 6. In Panel A, the average one-month return for the $Q4_netbuy_{jt}$ portfolio for long-term investors is larger than it is for the short-term group, but this difference is not statistically significant. In contrast, Panels B and C present evidence of better performance by long-term institutions. Both the alpha and twelve-month average return of the $Q4_netbuy_{jt}$ of long-term institutions is larger, although this difference is not statistically significant. Moreover, the $Q1_netbuy_{jt}$ ($Q4 - Q1_netbuy_{jt}$) portfolio for long-term institutions has statistically significant worse (better) alpha compared with the one for short-termers (Panel B). Besides, the $Q1_netbuy_{jt}$ portfolio of long-termers have significantly worse performance than for short-term investors (Panel C). In summary, long-term investors have an edge over short-term ones in a one-year investment period, and this is due in a larger degree to the stocks they tend to sell more ($Q1_netbuy_{jt}$) and, to a lesser extent, to those they tend to buy more ($Q4_netbuy_{jt}$).

¹² Larger trading positions of long-term institutions, such as pension and mutual funds, are more complicated to execute economically, because of the larger price impact.

¹³ Results on the average trading costs (equations 1 and 2) and the net profit 1 measure (equation 4) are omitted for the sake of brevity but are available upon request from the authors.

5. Conclusions

This paper investigates the differential performance of foreigners, institutions, and individuals in the Colombian stock market. The results consistently show that institutions are the better performers, both in terms of trade execution as well as in terms of returns in different horizons: the very short term (daily), a month, and a year. This advantage can be related to having better information, as detected in increasing adverse selection components of the bid-ask spread on the stock days that institutions are more actively trading. This is consistent with two notions: First, institutions are more professional, more savvy, and better-informed investors than individuals. Second, locals are better informed than foreigners, which is one of the main explanations of the home bias puzzle. As expected, in the short term, institutions have a positive performance particularly in the largest stocks and for both short-term (e.g., brokerage firms) and long-term (e.g., funds) institutions. Interestingly, institutions also do better in long-term performance, mostly because of the stocks they avoid or sell than for the stocks they buy or hold, and this is mostly driven by long-term institutions.

Foreigners are in second place. They appear to be the worst trade executors, which is already reported in the literature. However, they appear to have an edge over local individuals in terms of returns measured over a month and a year. In addition, they do better than individuals in two daily performance measures driven by market timing. Again, this shows that institutions do better than individuals, since foreigners are international institutional investors.

The results here presented reconcile some the contradictory findings in the literature. The question of who has the best performance in an emerging market can be answered differently depending on the context. Foreigners are clearly the worst executors of trades on a day-by-day

basis. Moreover, compared to the performance of institutions, locals are clearly better than foreigners. When comparing the two groups of locals, institutions are unambiguously better. However, if we pooled together the two types of local investors, foreigners might prove to have better performance, at least in some measures, simply because local individuals conduct most of the transactions.

The most important question to be explored in this line of research is the nature of informational advantage among the three types of investors. Is the local institution advantage firm specific? Country macro specific? Are foreigners better informed in terms of international financial and macro variables? Do individual or short-term institutions possess any advantage with respect to speculative stocks? We leave those questions for future research.

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AUTHOR (2015)

Table 1. Summary statistics of trading activity by type of investor and size quartile.

Average trading size (USD thousands)									
Size Quartile:	Total of operations (thousands)	Total traded value (USD millions)							
			Total of operations (thousands)	Total traded value (USD millions)	Mean	t-test > Foreigners		t-test > Local Individuals	
	Full Sample		Local Institutions						
Total	9,289	238,000	3,189	125,000	35.71	9.283	***	66.109	***
4	4,540	139,000	1,642	75,100	43.91	7.581	***	44.591	***
3	2,195	56,900	768	30,000	38.62	5.631	***	46.559	***
2	1,235	21,500	414	11,300	34.76	3.008	***	19.136	***
1	1,320	20,800	366	8,370	21.33	0.532		20.848	***

Size Quartile	Average trading Size (in USD thousands)							
	Total of operations (thousands)	Total traded value (USD millions)	Mean	t-test > Local Individuals	Total of operations (thousands)	Total traded value (USD millions)	Average trading Size (USD thousands)	
	Foreigners				Local Individuals			
Total	788	27,200	35.31	30.66 ***	5,312	86,000	14.86	
4	342	13,100	39.88	27.90 ***	2,556	50,600	17.99	
3	325	11,000	35.04	18.67 ***	1,102	15,800	15.04	
2	62	1,360	29.24	6.60 ***	759	8,860	11.44	
1	58	1,720	24.98	5.38 ***	896	10,700	8.63	

This table reports summary statistics of trading activity in the Colombian Stock Exchange (BVC) from January 1, 2007, to May 30, 2014, by different types of investors: summarized by quartiles market capitalizations, based on data provided by the BVC. A sample two-tailed t-test is presented to compare on the equality of average trading sizes among types of investors. *, **, *** Statistically significant at the 10%, 5%, and 1% level, respectively.

Table 2. Average trading cost in buys and sells by type of investor and size quartile.

Panel A

		Buys			Sales			Buys			Sales			Buys			Sales		
		N	Mean		N	Mean		N	Mean		N	Mean		N	Mean		N	Mean	
Quartile	Average trades per day	Local Institutions						Foreigners						Local Individuals					
Total	360	47,200	-0.01%	***	46,257	-0.03%	***	27,231	0.09%	***	23,526	0.07%	***	51,721	-0.10%	***	53,037	-0.02%	***
4	198	11,760	-0.02%	***	11,732	-0.02%	***	8,863	0.08%	***	7,801	0.06%	***	12,109	-0.07%	***	12,206	-0.02%	***
3	66	15,367	-0.03%	***	15,294	-0.04%	***	10,681	0.09%	***	9,156	0.06%	***	16,400	-0.09%	***	16,587	-0.04%	***
2	53	9,517	0.02%	***	9,138	-0.03%	***	3,765	0.08%	***	3,126	0.06%	***	10,446	-0.11%	***	10,941	-0.01%	**
1	44	10,556	0.01%		10,093	-0.04%	***	3,922	0.13%	***	3,443	0.10%	***	12,766	-0.13%	***	13,303	-0.01%	**

Panel B

Quartile	Average trades per day	Buys				Sells				Buys				Sells	
		Wilcoxon Test >		Wilcoxon Test >		Wilcoxon Test >		Wilcoxon Test >		Wilcoxon Test >		Wilcoxon Test >			
		Foreigners		Individuals		Foreigners		Individuals		Individuals		Individuals			
		Local Institutions								Foreigners					
Total	360	-34.17	***	29.29	***	-25.37	***	-10.33	***	50.62	***	23.40	***		
4	198	-21.66	***	14.83	***	-14.04	***	-2.28	**	29.15	***	14.29	***		
3	66	-26.09	***	13.43	***	-19.11	***	-6.45	***	35.22	***	16.92	***		
2	53	-7.33	***	14.23	***	-7.25	***	-5.89	***	14.90	***	4.54	***		
1	44	-8.62	***	16.82	***	-7.81	***	-5.67	***	18.07	***	8.67	***		

This table presents the results of average of daily trading cost in both buys (1) and sells (2), measured against the daily value-weighted average price, for the three types of investors, the full sample, and four quartiles of size. For each measure, the number of observations N, the mean, and statistical significance are reported. Based on data provided by the Colombian Stock Exchange (BVC) from January 1, 2007, to May 30, 2014 *, **, *** Statistically significant at the 10%, 5%, and 1% level, respectively.

Table 3. Daily performance measure by type of investor and size quartile.

		N	Mean			Wilcoxon Test > Foreigners			Wilcoxon Test > Individuals			N	Mean			Wilcoxon Test > Individuals	N	Mean		
Quartile	Average trades per day	Local Institutions								Foreigners				Local Individuals						
Total	360	52,100	0.01%			3.17	***		9.64	***		52,100	0.00%			9.02	***	52,100	-0.03%	
4	198	12,159	0.08%	***		2.65	***		7.87	***		12159	0.03%	**		6.04	***	12,159	-0.10%	***
3	66	16420	0.00%			-2.27	**		2.71	***		16,420	0.05%	***		5.34	***	16420	-0.04%	***
2	53	10,670	-0.08%			4.18	***		5.84	***		10,670	-0.11%			4.58	***	10,670	0.06%	
1	44	12,851	0.03%			2.39	**		3.32	***		12,851	-0.02%			2.13	**	12,851	-0.03%	

This table presents the results of the simple daily performance measure (3), for the three types of investors, the full sample, and four quartiles of size. The number of observations N, the mean, and statistical significance are reported. Based on data provided by the Colombian Stock Exchange (BVC) from January 1, 2007, to May 30, 2014. *, **, *** Statistically significant at the 10%, 5%, and 1% level, respectively.

Table 4. Daily net profit measure 1 by type of investor and size quartile.

		N		Mean		Wilcoxon Test > Foreigners		Wilcoxon Test > Individuals		N		Mean		Wilcoxon Test > Individuals		N		Mean	
Quartile	Average trades per day	Local Institutions								Foreigners				Local Individuals					
Total	360	52,100	0.03%	***	1.74	*	11.80	***	51,204	0.03%	9.53	***	52,100	-0.07%	***				
4	198	12,159	0.05%	***	1.96	**	9.10	***	12,159	-0.01%	5.61	***	12,159	-0.10%	***				
3	66	16,420	0.01%		-3.22	***	4.79	***	15,524	0.08%	**	7.12	***	16,420	-0.06%	***			
2	53	10,670	0.06%	***	3.12	***	5.73	***	10,670	0.03%	3.87	***	10,670	-0.10%	***				
1	44	12,851	0.02%	*	1.92	*	4.16	***	12,851	0.03%	2.25	**	12,851	-0.03%	**				

This table presents the results of the daily net profit measure 1 (4) , for the three types of investors, the full sample, and four quartiles of size. The number of observations N, the mean, and statistical significance are reported. Based on data provided by the Colombian Stock Exchange (BVC) from January 1, 2007, to May 30, 2014. *, **, *** Statistically significant at the 10%, 5%, and 1% level, respectively.

Table 5. Daily net profit measure 2 by type of investor and size quartile.

		N	Mean		Wilcoxon Test > Foreigners		Wilcoxon Test > Individuals		N	Mean		Wilcoxon Test > Individuals		N	Mean	
Quartile	Average trades per day	Local Institutions						Foreigners						Local Individuals		
Total	360	42330	0.04%	***	13.31	***	4.80	***	17535	-0.27%	***	-10.25	***	49012	-0.02%	**
4	44	11343	0.06%	***	9.49	***	5.26	***	6611	-0.23%	***	-5.44	***	12002	-0.06%	***
3	53	14417	0.04%	***	8.72	***	2.25	**	7566	-0.25%	***	-8.16	***	15862	0.01%	
2	66	7824	0.06%	**	2.68	***	2.29	**	1611	-0.25%		-1.23		9355	-0.05%	**
1	198	8746	0.00%		3.30	***	-0.28		1747	-0.50%	*	-3.95	***	11793	0.01%	

This table presents the results of the daily net profit measure 2 (5), for the three types of investors, the full sample, and four quartiles of size. The number of observations N, the mean, and statistical significance are reported. Based on data provided by the Colombian Stock Exchange (BVC) from January 1, 2007, to May 30, 2014. *, **, *** Statistically significant at the 10%, 5%, and 1% level, respectively.

Table 6. Performance of portfolios formed on monthly net buy quartiles by type of investor

1-month investor's portfolio performance														
		T-test					Alfa		Beta		Wald Test			
		N	Mean return	Standard error mean	vs. Foreigners	vs. Local Individuals	Coefficient	Standard error	Coefficient	Standard error	vs. Foreigners	vs. Local Individuals		
Type of Investor	Portfolio	Panel A					Panel B							
Local Institutions	$Q4_netbuy_{jt}$	87	0.940%	* 0.535%	1,085	1,006	0.21%	0.27%	0.845	0.052	***	2.070	1.150	
	$Q1_netbuy_{jt}$	87	0.154%	0.440%	-1.457	*	-1.648	**	-0.50%	0.29%	*	0.611	0.056	***
	$Q4 - Q1_netbuy_{jt}$	87	0.786%	* 0.439%	1.552	*	1.430	*	0.71%	0.42%	*	0.234	0.082	***
Foreigners	$Q4_netbuy_{jt}$	87	0.607%	0.490%		0.361	-0.11%	0.23%	0.790	0.044	***		1.370	
	$Q1_netbuy_{jt}$	87	0.707%	* 0.425%		-0.356	0.05%	0.25%	0.629	0.049	***		9.530	***
	$Q4 - Q1_netbuy_{jt}$	87	-0.100%	0.344%		0.436	-0.15%	0.34%	0.161	0.065	**		0.950	
Local Individuals	$Q4_netbuy_{jt}$	87	0.460%	0.618%			-0.28%	0.39%	0.877	0.075	***			
	$Q1_netbuy_{jt}$	87	0.828%	* 0.508%			0.10%	0.24%	0.819	0.046	***			
	$Q4 - Q1_netbuy_{jt}$	87	-0.368%	0.470%			-0.39%	0.47%	0.058	0.092				
COLCAP		87	0.785%	0.553%										

		12-month investor's portfolio performance							
		N	Mean return		Standard error mean	T-test			
						vs. Foreigners	vs. Local Individuals		
Type of Investor	Quartile	Panel C							
Local Institutions	$Q4_netbuy_{jt}$	76	0.618%	**	0.222%	0.204		1.954	**
	$Q1_netbuy_{jt}$	76	0.185%		0.147%	-1.975	**	-3.783	***
	$Q4 - Q1_netbuy_{jt}$	76	0.433%	**	0.152%	1.455	*	3.287	***
Foreigners	$Q4_netbuy_{jt}$	76	0.598%	**	0.207%			1.701	**
	$Q1_netbuy_{jt}$	76	0.444%	**	0.185%			-2.071	**
	$Q4 - Q1_netbuy_{jt}$	76	0.154%		0.134%			2.242	**
Local Individuals	$Q4_netbuy_{jt}$	76	0.390%	*	0.220%				
	$Q1_netbuy_{jt}$	76	0.705%	***	0.208%				
	$Q4 - Q1_netbuy_{jt}$	76	-0.315%	**	0.124%				
COLCAP		76	0.652%	***	0.194%				

This table describes the performance of the equally weighted portfolios formed by the monthly top ($Q4_netbuy_{jt}$) and bottom quartile ($Q1_netbuy_{jt}$) stocks by monthly net buy for each of the three type of investors: local institutions, foreigners, and local individuals. The performance of a portfolio long in the top quartile and short in the bottom one, $Q4 - Q1_netbuy_{jt}$ is also reported. In Panels A and B, the performance is computed from monthly returns of the portfolios rebalanced at the end of the month. In Panel C, the performance is computed without rebalancing, as the effective monthly return in a twelve-month period. The paired two-tailed t-test in Panels A and C verify whether the mean return of the portfolio in the row is higher or lower than the mean return of the portfolio in the column. Panels A and C report the summary statistics and comparisons between types of investors using a two-tailed t-test. Panel B reports the results of three-equation SUR models, one for each class of portfolio, regressing monthly excess returns against the market index excess returns, using the TIB (Tasa interbancaria) as the risk-free rate. Panel B also reports Wald test for the difference between the alphas in the SUR models. Based on data provided by the Colombian Stock Exchange (BVC) from January 1, 2007, to May 30, 2014. *, **, *** Statistically significant at the 10%, 5%, and 1% level, respectively.

Table 7. Regressing the implied spread and its adverse selection component against trading intensity by type of investor

Dependent variable:	<i>Implied_spread_{it}</i>		<i>Adv_selec_comp_{it}</i>		<i>Implied_spread_{it}</i>		<i>Adv_selec_comp_{it}</i>		<i>Implied_spread_{it}</i>		<i>Adv_selec_comp_{it}</i>	
<i>Volat_{it}</i>	4.107	***	1.100	***	4.105	***	1.077	***	4.004	***	1.104	***
<i>Log_num_op_{it}</i>	-0.061	***	-0.015	***	-0.065	***	-0.100	***	-0.056	***	-0.012	***
<i>return_{it}</i>	0.014		-0.077		0.031		-0.062		0.022		-0.080	
<i>Log_Price_{it}</i>	-0.028	***	-0.001		-0.029	***	-0.002	**	-0.030	***	-0.003	***
<i>Log_Avg_trade_Size_{it}</i>	-0.035	***	-0.006	**	-0.034	***	-0.006	**	-0.030	***	-0.004	***
<i>Log_ OrderImbal _{it}</i>	-0.005	**	0.002	**								
<i>Log_ OrderImbal × Share_Foreigner_{it}</i>					-0.027	**	-0.014	***				
<i>Log_ OrderImbal × Share_Institutional_{it}</i>					-0.004		0.006	**				
<i>Log_ OrderImbal × Share_Individual_{it}</i>					-0.001		0.004	*				
<i>Log_ Netbuy_Foreigner _{it}</i>									-0.008	***	-0.004	***
<i>Log_ Netbuy_Institutional _{it}</i>									0.000		0.002	**
<i>Log_ Netbuy_Individuals _{it}</i>									-0.005		-0.002	*
$\bar{R}^2$	0.248		0.111		0.250		0.122		0.245		0.131	
No. Observations	2,121		2,121		2,121		2,121		2,013		2,013	
<i>Wald test</i>												
<i>Foreigner = Institutional</i>							18.080	***			25.370	***
<i>Foreigner = Individual</i>							17.010	**			3.260	*
<i>Institutional = Individual</i>							0.390				6.330	**

This table reports the results of panel data regressions (7) and (8) of the implied spread and its adverse selection component against trading intensity by type of investor. Coefficients appear multiplied by 1.000. The model regresses the estimated implied spread, estimated as $(\alpha + \beta)$ and the adverse selection component (α), both scaled by the average trading price in the stock-day against measures of trading intensity by type of investor, and known determinants of the bid-ask spread. Implied spreads and adverse selection components (α , β) are estimated in a stock-day basis following the MRR model that uses a GMM specification [6] based on transaction prices. The last three rows presents the result of a Wald test on the difference between two coefficients of trading intensity. The panel data model is estimated with fixed effects and the Driscoll & Kraay covariance matrix estimator (Driscoll & Kraay, 1998). Based on transaction data taken from Bloomberg data and daily data by the Colombian Stock Exchange (BVC) from August 23, 2010, to August 31, 2012. *, **, *** Statistically significant at the 10%, 5%, and 1% level, respectively.

Table 8. Summary statistics of trading by type of local institutions.

Type of Institutional Investor	Panel A					Panel B					Panel C
	Total traded value (USD millions)					Number of operations (thousands)					Average trading size (USD thousands)
Quartile:	Total	4	3	2	1	Total	4	3	2	1	Total
Brokerage Firms	41,040.0	30,400.0	5,500.0	3,190.0	1,950.0	1,202.5	736.3	193.9	170.6	101.7	34.1
Family Offices	35,180.0	18,100.0	8,990.0	4,170.0	3,920.0	1,110.3	479.4	299.6	144.6	186.8	31.7
Pension Funds	28,960.0	16,400.0	9,700.0	2,080.0	780.0	356.2	198.3	112.5	28.8	16.6	81.3
Mutual Funds	14,302.0	8,000.0	3,980.0	1,360.0	962.0	395.6	183.2	115.1	52.5	44.8	36.2
Insurance Companies	2,462.3	1,330.0	752.0	296.0	84.3	47.0	22.8	14.4	6.9	2.9	52.4
Investment Companies	1,288.0	451.0	374.0	343.0	120.0	29.1	10.5	8.9	3.9	5.9	44.2
Investment Trusts	985.0	262.0	195.0	103.0	425.0	28.6	8.0	9.1	6.4	5.1	34.5
Banks	348.8	169.0	3.4	2.4	174.0	3.6	1.9	0.1	0.1	1.6	95.9
Financial Corporations	105.3	0.4	18.9	0.0	86.0	0.1	0.0	0.1	0.0	0.0	706.8
Credit Union	48.7	29.1	9.9	5.9	3.8	2.5	1.3	0.6	0.3	0.3	19.5
Factoring Companies	18.2	14.0	2.3	1.9		0.4	0.3	0.1	0.1	0.0	41.5
Total	124,738	75,156	29,526	11,552	8,505	3,176.0	1,641.8	754.4	414.1	365.6	39.3

This table reports the summary statistics for trading activity for each type of investor, the four quartiles of size, and full sample.

Table 9. Daily performance of short-term vs. long-term local institutions.

<u>Panel A: Simple Daily Performance Measure</u>								
Quartile	Average trades per day	N	Mean	ST>LT		N	Mean	
		Short-Term Local Institutions			Long-Term Local Institutions			
Total	111	52,100	-0.029%		-1.759	*	52,100	0.008%
4	62	12,159	-0.007%		-2.890	**	12,159	0.053% ***
3	21	16,420	-0.034%	***	-3.365	***	16,420	0.023% **
2	17	10,670	-0.083%		-0.056		10,670	-0.078%
1	11	12,851	0.001%		-0.604		12,851	0.018%

<u>Panel B: Net Profit Measure 2</u>								
Quartile	Average trades per day	N	Mean	ST>LT		N	Mean	
		Short-Term Local Institutions			Long-Term Local Institutions			
Total	111	38,122	0.063%	***	1.118		24,771	0.034%
4	62	10,760	0.063%	***	-0.661		8,326	0.105% **
3	21	13,094	0.042%	**	0.090		9,953	0.035%
2	17	6,628	0.158%	***	2.262	**	3,817	-0.134%
1	11	7,640	0.018%		0.178		2,675	0.043%

This table presents the results of the daily performance measures [3] (simple measure) and [5] (net profit measure 2), for short-term and long-term local institutions, the full sample, and four quartiles of size. The ST>LT columns present the result of a t-test on the difference between the two means performance measures. The number of observations N, the mean, and statistical significance are reported. Based on data provided by the Colombian Stock Exchange (BVC) from January 1, 2007, to May 30, 2014. *, **, *** Statistically significant at the 10%, 5%, and 1% level, respectively.

Table 10. Investor's portfolio performance of short-term vs. long-term local institutions.

		1-month investor's portfolio performance									
		Panel A				Panel B					
Type of Investor	Portfolio	N	Mean return	Standard error mean	ST>LT	Alpha		Beta		ST>LT	
						Coefficient	Standard error	Coefficien t	Standard error		
Short-Term	$Q4_netbuy_{jt}$	87	0.759%	0.510%	-0.0847	0.049%	0.275%	0.779	0.053	***	2.840
Local Institutions	$Q1_netbuy_{jt}$	87	0.527%	0.516%	0.5533	-0.195%	0.255%	0.815	0.050	***	35.880
	$Q4 - Q1_netbuy_{jt}$	87	0.232%	0.354%	-0.3723	0.244%	0.352%	-0.036	0.068		18.190
Long-Term	$Q4_netbuy_{jt}$	87	0.790%	0.553%		0.041%	0.252%	0.895	0.049	***	
Local Institutions	$Q1_netbuy_{jt}$	87	0.312%	0.330%		-0.282%	0.226%	0.437	0.044	***	
	$Q4 - Q1_netbuy_{jt}$	87	0.478%	0.448%		0.323%	0.368%	0.458	0.071	***	

Panel C: 12-month investor's portfolio performance

Type of Investor	Portfolio	Portfolio			T-test	
		N	Mean return	Standard error mean	ST>LT	
Short-Term	$Q4_netbuy_{jt}$	76	0.520% **	0.201%	0.068	
Local Institutions	$Q1_netbuy_{jt}$	76	0.512% **	0.211%	1.837 **	
	$Q4 - Q1_netbuy_{jt}$	76	0.008%	0.105%	-1.229	
Long-Term	$Q4_netbuy_{jt}$	76	0.513% **	0.226%		
Local Institutions	$Q1_netbuy_{jt}$	76	0.237% **	0.112%		
	$Q4 - Q1_netbuy_{jt}$	76	0.277% *	0.163%		

This table describes the performance of the equally weighted portfolios formed by the monthly top ($Q4_netbuy_{jt}$) and bottom quartile ($Q1_netbuy_{jt}$) stocks by monthly net buy for short-term and long-term institutional investors. The performance of a portfolio long in the top quartile and short in the bottom one, $Q4 - Q1_netbuy_{jt}$ is also reported. In Panels A and B, the performance is computed from monthly returns of the portfolios rebalanced at the end of the month. In Panel C the performance is computed without rebalancing, as the effective monthly return in a twelve-month period. The paired two-tailed t-test in Panels A and C verify whether the mean return of the portfolio in the row is higher or lower than the mean return of the portfolio in the column. Panels A and C report the summary statistics and comparisons between types of investors using a two-tailed t-test (ST>LT columns). Panel B reports the results of two-equation SUR models, one for each class of portfolio, regressing monthly excess returns against the market index excess returns, using the TIB (Tasa interbancaria) as the risk-free rate. Panel B also reports Wald test results for the difference between the two alphas in the SUR models. Based on data provided by the Colombian Stock Exchange (BVC) from January 1, 2007, to May 30, 2014. *, **, *** Statistical significant at the 10%, 5%, and 1%, respectively.